Track 2: Internet of Things (IoT)

This track focuses on low-cost sensing, communication, and control systems that bring real-time information and automation to farms and post-harvest operations. Solutions should be rugged, energy-conscious, and simple enough for farmers to install and maintain.

Problem Title 1: Smart Irrigation and Water Management

A. Problem Description

Many farms irrigate on fixed schedules or by visual judgement, which can result in over-irrigation, waterlogging, wasted energy, or crop stress during dry periods. Water scarcity makes efficient use increasingly important, particularly where groundwater is limited.

Sensors that measure soil moisture and local conditions, combined with automated valve or pump control, can supply water when and where the crop needs it.

B. Challenges

1. Sensor accuracy, calibration, and drift in different soil types and over a growing season.
2. Durability of electronics and sensors against moisture, dust, heat, and rodents.
3. Reliable wireless communication across large or obstructed fields.
4. Power supply for remote nodes and pumps, including low-power design.
5. Keeping the system affordable and easy for farmers to configure and override manually.

Problem Title 2: Smart Agricultural Leaf Drying System

A. Problem Description

Leaves such as tobacco and tea must be dried under controlled conditions to reach the required moisture level while retaining colour, aroma, and quality. Traditional sun or manual drying is affected by weather and depends on operator experience, causing uneven drying, mould growth, and quality loss.

A monitored and regulated drying chamber can keep conditions within suitable ranges and reduce the need for constant supervision.

B. Challenges

1. Different leaf types require different temperature, humidity, and duration profiles.
2. Achieving uniform airflow and temperature across the drying chamber.
3. Estimating moisture content or drying progress without destructive testing.
4. Controlling energy consumption and heating cost.
5. Ensuring safe operation with appropriate limits and fail-safes for heating elements.

Problem Title 3: Smart Agricultural Storage and Spoilage Prevention

A. Problem Description

Post-harvest losses occur when grains, pulses, fruits, and vegetables are stored under unsuitable temperature and humidity. Moisture promotes fungal growth, while heat accelerates ripening and decay. In many small godowns and farm stores, conditions are checked only occasionally, if at all.

Continuous monitoring with timely alerts can help owners act before losses become severe.

B. Challenges

1. Sensor placement and representativeness within large or stacked storage spaces.
2. Defining suitable thresholds for different commodities.
3. Maintaining connectivity in remote or poorly connected storage sites.
4. Power backup and low-power operation during outages.
5. Preventing false alerts and ensuring alerts reach responsible persons promptly.

Problem Title 4: Smart Coconut Farming and Plantation Monitoring

A. Problem Description

Coconut plantations are often large, spread across uneven terrain, and require regular irrigation, particularly in dry months. Owners may have limited visibility of conditions in distant parts of the plantation and of water sources such as wells and tanks.

Distributed sensing can show which areas are dry, whether water sources are adequate, and when attention is needed.

B. Challenges

1. Deploying sensors over large, tree-covered areas with dependable communication coverage.
2. Placing soil sensors appropriately around tall palms with deep root zones.
3. Protecting equipment from heat, rain, insects, and animals.
4. Powering remote nodes for long periods at low cost.
5. Presenting data in a simple form that supports irrigation decisions.

Problem Title 5: Development of a Low-Cost, Lightweight Milk Chilling Can for Small-Scale Dairy Farmers

A. Problem Description

Milk is highly perishable and can deteriorate quickly when stored at unsuitable temperatures. Small-scale dairy farmers in rural and remote areas often lack access to reliable milk-cooling facilities and electricity. During milk collection and transportation, exposure to high temperatures can reduce milk quality and cause financial losses.

Develop a low-cost, lightweight, and portable milk-chilling can that helps small-scale dairy farmers maintain milk at an appropriate temperature for several hours. The design should use effective insulation and an affordable cooling approach, such as reusable ice packs or phase-change materials, to reduce dependence on continuous electricity. The system may also include temperature monitoring and alerts to help farmers track storage conditions and identify when the milk temperature rises beyond the defined safe range.

The solution should prioritize food safety, portability, durability, ease of cleaning, and affordability for rural dairy farmers.

B. Challenges

1. Maintaining an appropriate milk temperature for several hours under changing environmental conditions.
2. Designing effective insulation and an affordable cooling mechanism.
3. Monitoring milk temperature accurately and providing timely alerts.
4. Reducing energy requirements and supporting use in areas with unreliable electricity.
5. Ensuring food-safe materials, leak prevention, and hygienic cleaning.
6. Keeping the can lightweight, durable, easy to transport, and affordable for small-scale farmers.

Problem Title 6: Smart Farm Security and Crop Protection

A. Problem Description

Farms suffer from theft of produce and equipment, trespassing, and crop damage by animals. Constant manual watching is tiring and often impractical, especially at night and on remote plots.

A sensor-based alert system can notify farmers of suspicious activity, allowing a timely and proportionate response, while non-lethal deterrents protect crops without harming wildlife.

B. Challenges

1. Distinguishing real threats from false triggers caused by wind, small animals, or vegetation.
2. Covering wide, open areas at reasonable cost.
3. Operating at night and in rain, heat, and dust.
4. Maintaining power and connectivity in remote locations.
5. Using deterrents that are humane, legal, and safe for people and wildlife.